

Robust And Energy Efficient Controls For A Two-Level System

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Introduction

- Quantum technologies rely on **precise control** of quantum systems.
- Realistic implementations often face **parameter uncertainties**, which compromise control fidelity.
- Goal: Develop **robust control protocols** that maintain performance under allowed limit of uncertainties
- Minimize the **energy cost** associated with the generation of the control protocols

Formulation

we define action S formulated based on the ideas of control theory[1] as :

$$S = \Phi(|\psi_\tau\rangle) + \int_0^\tau L(|\psi(t)\rangle, u(t), t) dt$$

The lagrangian can be expressed as:

$$\mathcal{L} = F_0(\vec{u}(t), t) + \Re \left[\langle \chi(t) | \dot{\psi}(t) \rangle + i \langle \chi(t) | H(t) | \psi(t) \rangle \right]$$

*here $|\psi\rangle, |\chi\rangle$ are the state and costate of the system, $F_0(\vec{u}(t), t)$ represent the running cost and $\Phi(|\psi(\tau)\rangle)$ represent the terminal cost for the setup.

The optimal solution minimizes the action S. This optimality condition implies $\delta S = 0$. Solving this relation yields the dynamics of $|\psi\rangle, |\chi\rangle$ and the boundary conditions for the optimal trajectory.

$$\begin{aligned} * |\dot{\psi}(t)\rangle &= -i H(t) |\psi(t)\rangle. & * \langle \dot{\chi}(t) | &= i \langle \chi(t) | H(t) \\ * |\chi(\tau)\rangle &= -2 \frac{\partial \Phi}{\partial \langle \psi(\tau) |}. & * \frac{\partial F_0}{\partial u} &= \Im \langle \chi | \frac{\partial H}{\partial u} | \psi \rangle \end{aligned}$$

The Pontryagin Hamiltonian for this system can be expressed as :

$$H_p = \Im \langle \chi | H | \psi \rangle - F_0$$

Algorithm

- Choose an initial control $\vec{u}(t)$
- Select an η value from the discretized set ie from $\eta \in \{-0.15, \dots, 0.15\}$, Find $|\psi\rangle, |\chi\rangle$ based on $H = H_0 + (1 + \eta)H_c$
- Find H_p for each system and update the control sequentially for each η with the following updation rule:

$$\vec{u} = \vec{u} + \epsilon \left(\frac{\partial H_p}{\partial \vec{u}} \right)$$

- Repeat 2 and 3 till the solution converges

Conclusion

We conclude our study by confirming the potential for achieving controls that can perform very well in spite of the uncertainty in the control amplitude, with a trade-off in energy consumption, and demonstrating better performances compared to previously known methods (Shortcut-to-adiabaticity).

System

We consider a two-level quantum system governed by a Landau-Zener model Hamiltonian H_0 .

$$H_0(t) = \Delta \sigma_x + \omega(t) \sigma_z, \quad \omega(t) = -10 + 20 \left(\frac{t}{\tau} \right)$$

The goal is to design a control protocol that transfers the system from the initial ground state to the final ground state with the help of the external control H_c within a fixed time denoted by τ .

$$H_c(t) = \sum_{\{x,y,z\}} \sigma_i u_i(t)$$

The total hamiltonian for the control problem can be represented as :

$$H(t) = H_0(t) + \omega_i H_c(t), \quad \omega_i = \sqrt{\Delta^2 + \omega^2(0)}$$

The energy associated with the control is expressed as the sum of squared control fields scaled by a factor.

$$C = \frac{\omega_i}{2} \int_0^\tau \left(u_x^2(t) + u_y^2(t) + u_z^2(t) \right) dt$$

The objective is to minimize this energy cost while steering the system from the initial to the target state, while ensuring fidelity.

Robustness

In real experiments, control fields can have small calibration errors or fluctuations can alter the system Hamiltonian.

To design **robust** controls, we account for these uncertainties directly in the optimization.

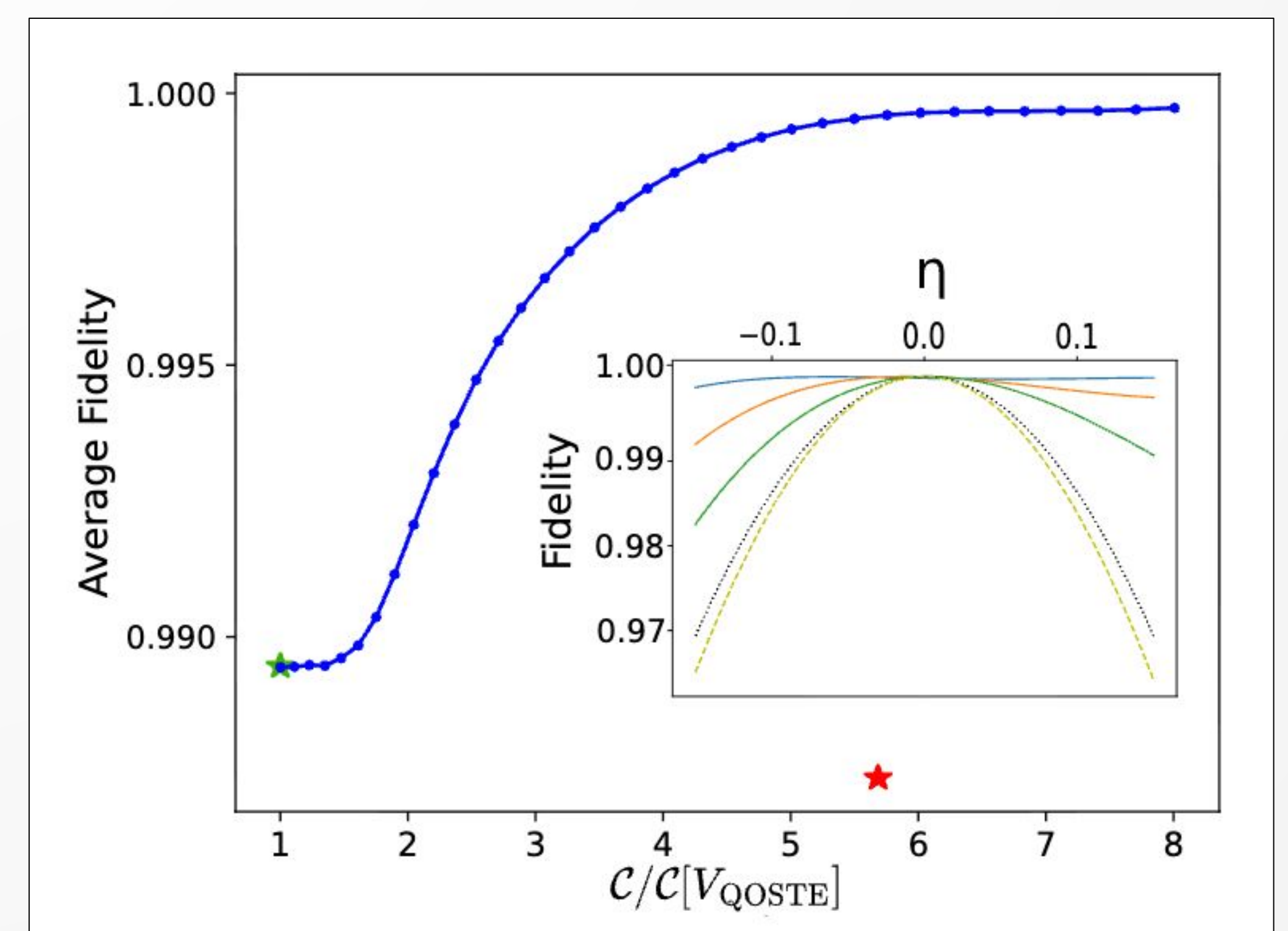
$$H = H_0 + (1 + \eta)H_c, \quad \eta \in [-0.15, 0.15]$$

The idea is to find a protocol that can yield better result under the variation of η within the given limits.

We consider an ensemble of systems for a set of $\eta \in \{-0.15, 0.15\}$ and consider the collective figure of merit to minimize as follows:

$$J = \frac{1}{N_\eta} \sum_{\eta} |\langle \psi_g | \psi_\eta(\tau) \rangle|^2 - \left(\frac{\omega_i}{2} \right) \int_0^{t_f} (\vec{u})^2 dt.$$

Result



The optimal solution minimizing the energy cost can be solved analytically without optimizing the robustness. The associated energy cost and average fidelity corresponds to the green star. Then, in order to optimize the robustness, we use the gradient-ascent algorithm described in the left box allowing for an increasing energy cost of the control. The obtained curve corresponds to the blue dots. The red star corresponds to the so-called Shortcut-to-adiabaticity, a widely used method the speed-up adiabatic trajectories. We see that our results outperform usual methods. We see also a trade-off between energy cost and robustness[2]. The average fidelity in y axis is expressed as $F_{avg} = \frac{1}{N_\eta} \sum |\langle \psi_g | \psi_\eta(\tau) \rangle|^2$. The energy in x axis is expressed as multiple of the energy cost obtained by analytical method

Acknowledgements: This work was supported by funding from the French National Research Agency (ANR) under grant ANR-23-CPJ1-0030-01.

References:

- [1] U. Boscain, M. Sigalotti, and D. Sugny, Introduction to the Pontryagin Maximum Principle for Quantum Optimal Control, PRX Quantum 2, 030203 (2021)
- [2] C. L. Latune, M. B. Puthuvedu Shebeek, D. Sugny, and S. Guérin. Energy Shortcut of Quantum Protocols by Optimal Control. arXiv preprint arXiv:2503.20130, 2025.